CHECKPOINT 1 · OCTOBER 20, 2026 · 15 MINUTES

Online Testing Application

Network Programming project · preliminary design and progress

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Objectives and main functions

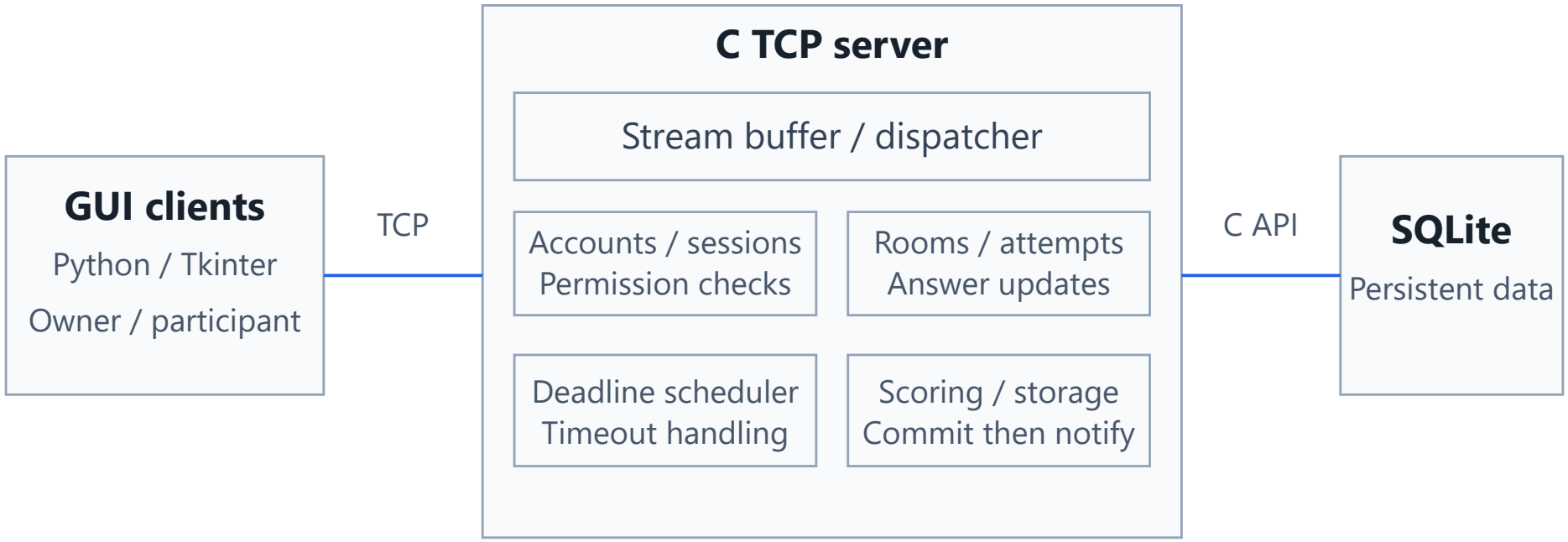
Test Mode

- Create room: question count and duration.
- View rooms; join before start.
- Owner starts/ends; users can revise answers.
- Submit or timeout → score and saved results.

Practice and shared features

- Choose count/time; practice and view score.
- Register/login and room permissions.
- Filter questions by topic and difficulty.
- GUI, attempt history, and score charts.

Tools and system architecture



POSIX sockets + pthreads; shared state and per-socket send locks. GCC, cJSON, SQLite, libsodium, Docker Compose.

Application-layer protocol

Field	Type	Rule
`type`	string	Command name, RESPONSE, or EVENT
`request_id`	positive integer	Request/response correlation
`event`	string	Required for server-push events
ok	boolean	Response success or failure
`data / error`	object	Payload, or error code + message

```
{"type":"ROOM_JOIN","request_id":42,"data":{"test_id":41}}\n
```

TCP reads are not messages

Two valid requests may arrive across only two `recv()` calls:

```
recv #1 — incomplete: keep in buffer
{"type":"ACCOUNT_

recv #2 — completes request 1, then contains request 2
GET","request_id":1,"data":{}}\n
{"type":"LOGOUT","request_id":2,"data":{}}\n
```

> Buffer → split at LF (0x0A) → parse JSON → validate fields. Maximum 64 KiB; oversized frames close the connection.

Commands and access rules

Group	Client requests (selected)	Server events
Account	REGISTER, LOGIN, LOGOUT	SESSION_EXPIRED
Room	ROOM_CREATE, ROOM_LIST, ROOM_JOIN ROOM_START, ROOM_END	ROOM_STATE TEST_STARTED
Attempt	QUESTIONS_GET, ANSWER_SET ATTEMPT_SUBMIT	ATTEMPT_FINISHED TEST_FINISHED
Results	ROOM_RESULTS_GET HISTORY_GET, STATS_GET	—

Every request receives RESPONSE with the same request_id. Events arrive independently. Permissions are checked per action.

Test and practice workflows

Test room lifecycle

NOT STARTED
Create · join

→

ONGOING
Answer · revise · submit

→

FINISHED
Score · results

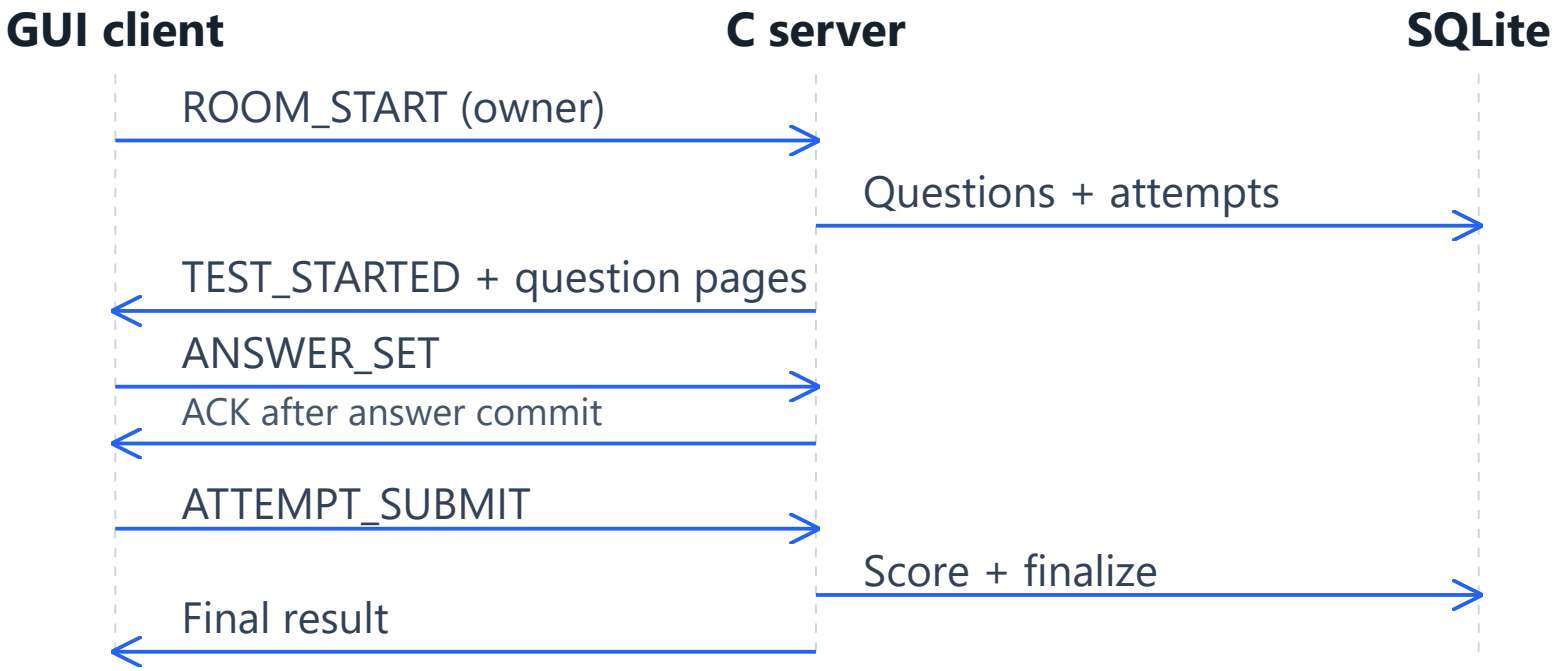
Test Mode

Owner controls start/end. Join before start. Server enforces the deadline.

Practice Mode

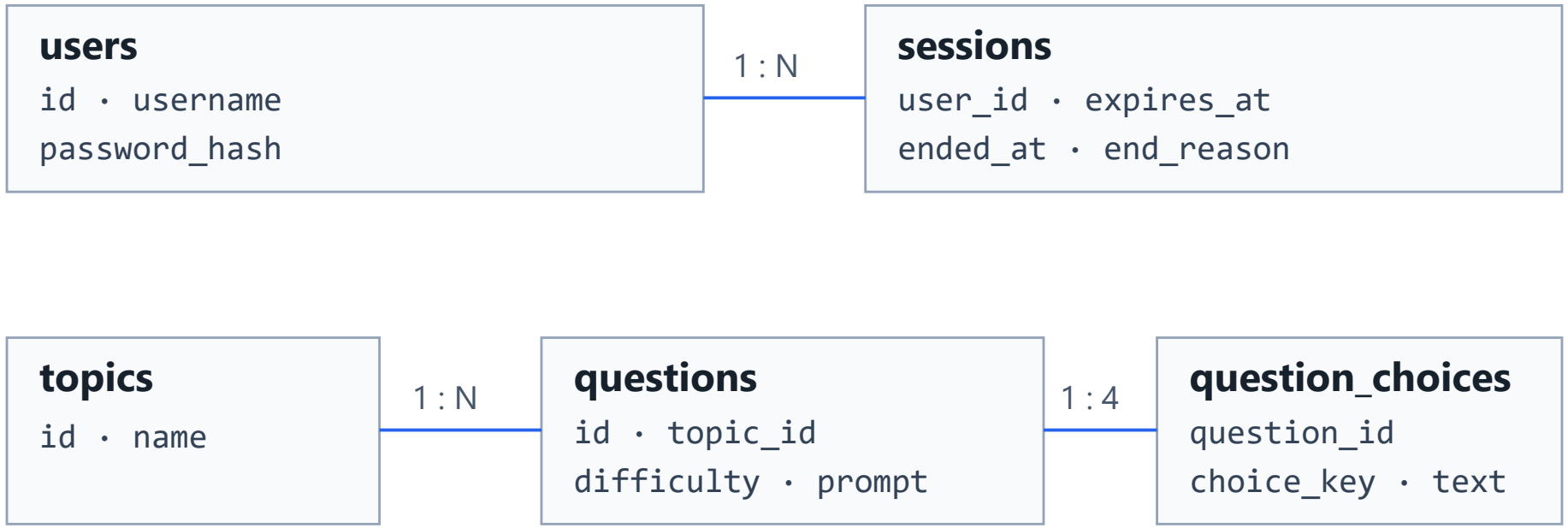
Choose count/time → answer → submit or timeout → score.

Test message sequence



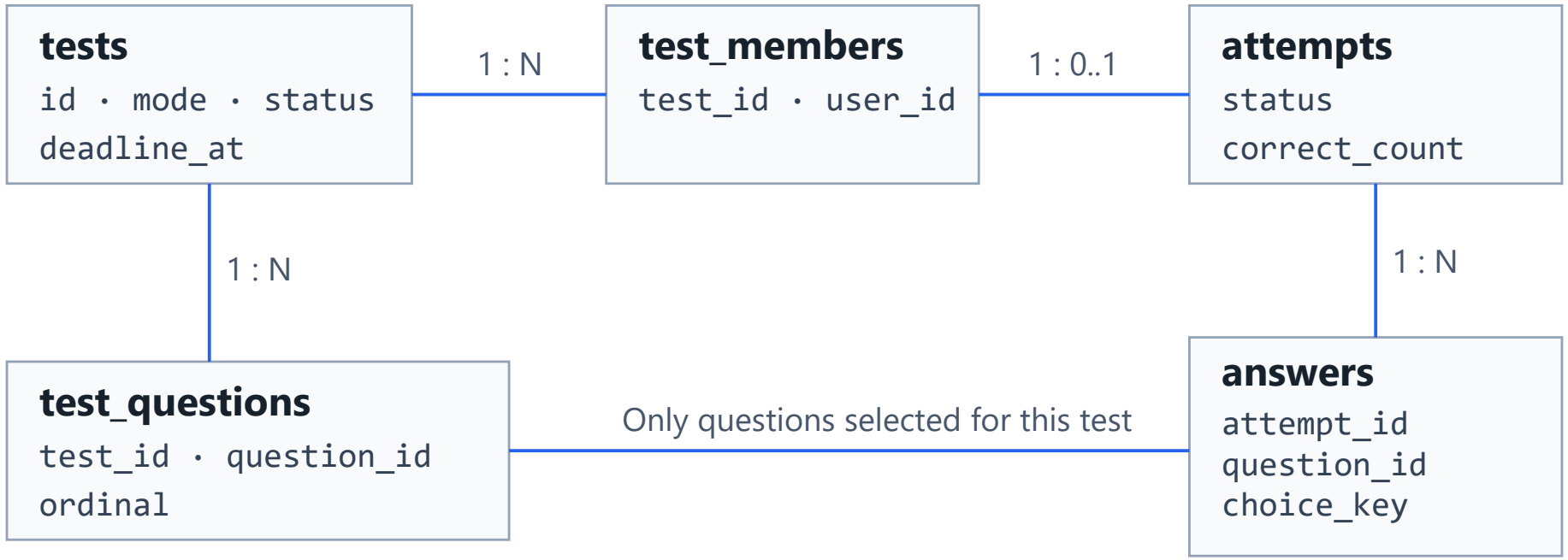
Timeout and owner-end use the same finalization path. Saved answers are committed before acknowledgement.

Accounts and question bank



Four choices, one correct key. Used questions stay immutable so stored results remain valid.

Shared test, individual attempts



Example: room 41 has attempts 11 and 12. Submitting attempt 11 leaves attempt 12 active. Practice has one attempt.

Question filters and statistics

Question bank

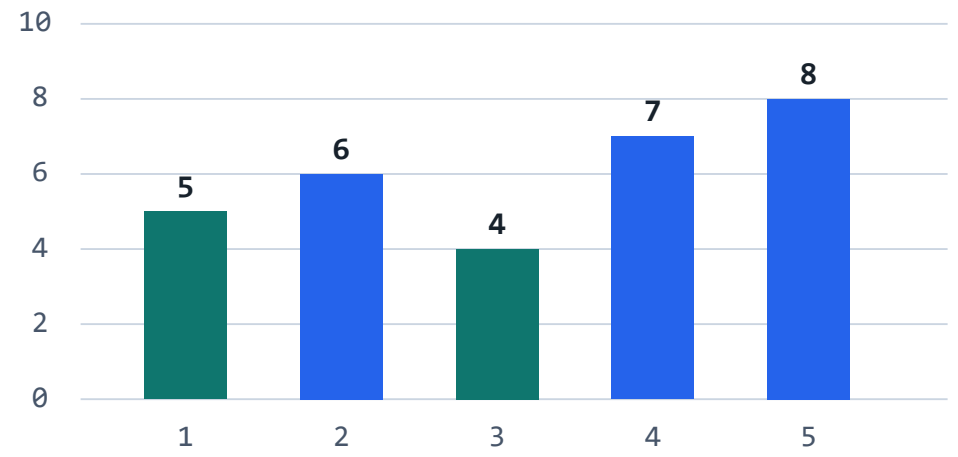
Filter by topic and difficulty; choose question count.

Store: questions, attempts, answers, scores, and activity logs.

The chart uses sample values only; no real user results exist yet.

Practice scores / 10

SAMPLE



Team responsibilities

Member	Student ID	Responsibility
Trần Hữu Đức	20225963	C sockets and stream I/O; room/attempt state, deadlines and scoring
Trần Anh Dũng Project lead	20226031	Architecture and message contracts; coordinate integration, review and demo
Trần Cao Phong	20226061	SQLite, accounts/sessions, permission checks and activity logging
Lê Diễm Hằng	20225970	Python TCP client, Tkinter UI, history/statistics and end-to-end tests

Completed design and next steps

Designed / checked

Flows, ERD, SQL schema, protocol; 12 schema checks

Next

C TCP framing → accounts → attempts and rooms

Not implemented

C application handlers and GUI

TCP + accounts

Framing, sessions

→

Test services

Practice, rooms, scoring

→

GUI + integration

History, charts, tests

Validate partial reads/writes, multiple clients, answer changes near deadlines, and early submission.